Jenkins

Class no:1

Jenkins

Jenkins is an open-source automation server that helps developers build, test, and deploy their software.

Think of it as the ultimate project coordinator for software development. Instead of a human having to manually compile code, run tests, and push updates to a server every time a change is made, Jenkins automates the entire sequence.

Here is a breakdown of how it works, why it's so popular, and where it fits in the development lifecycle.

The Jenkins will be installed default in **/Jenkins/var/lib/Jenkins**

Default port number: **port 8080**

The Core Concept: CI/CD

Jenkins is the industry standard tool for implementing **CI/CD**, which stands for:

- **Continuous Integration (CI):** Every time a developer commits new code to a repository (like GitHub or GitLab), Jenkins automatically triggers a "build." It compiles the code and runs automated tests to ensure the new changes didn't break anything.
- **Continuous Delivery/Deployment (CD):** If the build and tests pass, Jenkins can automatically deploy the updated application to a staging or production server.

How Jenkins Works (The Pipeline)

Jenkins uses a feature called **Pipelines**, which allows you to define your entire build process as code (usually written in a file called a Jenkinsfile).

A typical Jenkins pipeline follows these steps:

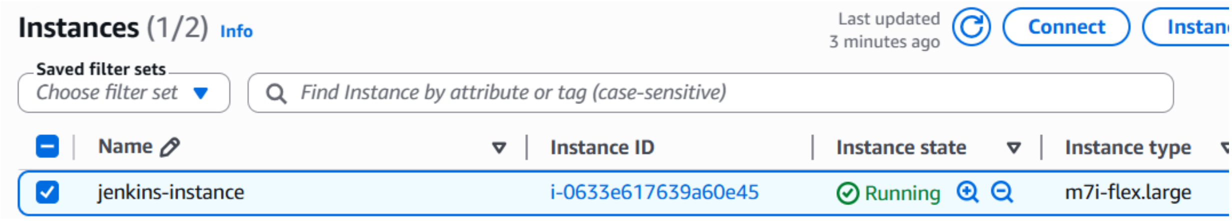
[Code Commit] ──> [1. Fetch Code] ──> [2. Build/Compile] ──> [3. Run Tests] ──> [4. Deploy]

1. **Trigger:** A developer pushes code to GitHub.
2. **Build:** Jenkins detects the push, grabs the code, and compiles it (e.g., turning Java code into a JAR file).
3. **Test:** Jenkins runs unit tests and integration tests. If a test fails, it alerts the team immediately.
4. **Deliver/Deploy:** If all tests pass, Jenkins packages the application (perhaps into a Docker container) and pushes it to your servers.

<u>Hands on : Launching and set up of Jenkins Via AWS EC2</u>

<u>Step no:1</u> Launch an EC2 instance (ubuntu) with a larger size to run Jenkins.

Choose default SG and keep All TCP enabled in it. (default port no. for Jenkins:8080)



<u>Step no:2</u> Jenkins requires java to be installed as prerequisites.

Go to Jenkins installation → Linux → Ubuntu

Installation of Java

Jenkins requires Java to run, yet not all Linux distributions include Java by default. Additionally, not all Java versions are compatible with Jenkins.

The Jenkins will be installed default in **/Jenkins/var/lib/Jenkins**

Default port number: **port 8080**

Copy and paste this code in the Aws console:

sudo apt update

sudo apt install fontconfig openjdk-21-jre

java -version

```
root@ip-172-31-14-119:/home/ubuntu# sudo apt update
sudo apt install fontconfig openjdk-21-jre
java -version
```

Installation of Jenkins

Copy and paste this code in the console after installing the java package.

Long Term Support release

A LTS (Long-Term Support) release is chosen every 12 weeks from the stream of regular releases as the stable release for that time period. It can be installed from the `debian-stable` apt repository.

```
sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
  https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]" \
  https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
  /etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install jenkins
```

```
root@ip-172-31-14-119:/home/ubuntu# sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
  https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]" \
  https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
  /etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install jenkins
```

Start the Jenkins server and check the status is running.

```
root@ip-172-31-14-119:/home/ubuntu# systemctl start jenkins
root@ip-172-31-14-119:/home/ubuntu# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service;
     Active: active (running) since Mon 2026-06-01 15:26:50 U
 Invocation: bce4554e4981420cb45ab7774a750cb8
```

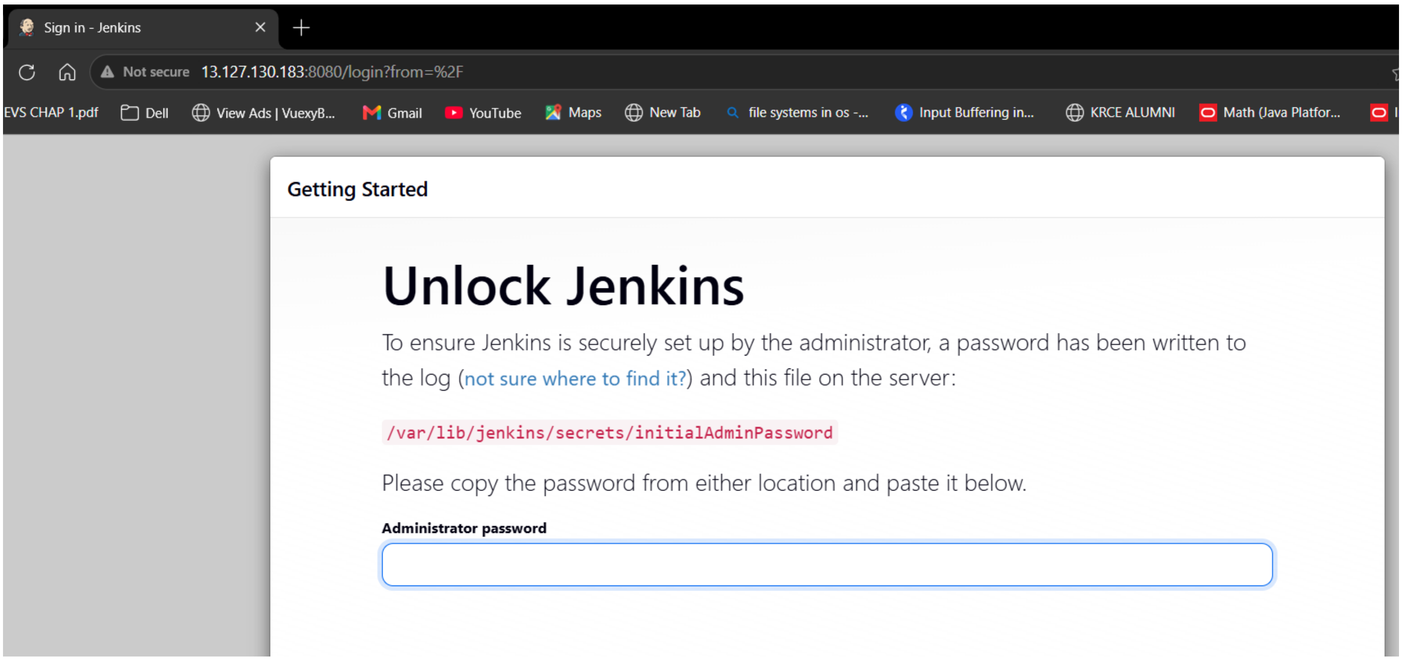
Step no:3 Copy the public IP of the instance:8080 (default port of Jenkins) and paste it in a private browser.

```
No VM guests are running outdated hypervisor
root@ip-172-31-14-119:/home/ubuntu#
```

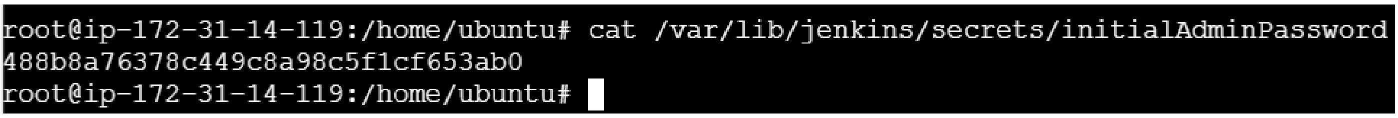
i-0633e617639a60e45 (jenkins-instance)

PublicIPs: 13.127.130.183   PrivateIPs: 172.31.14.119

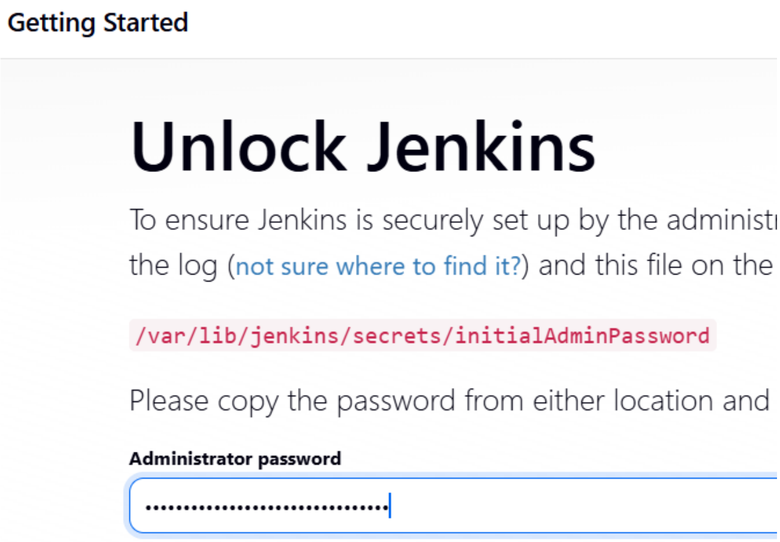
It will provide login page for Jenkins and the path where the password is located for login. Copy the path.



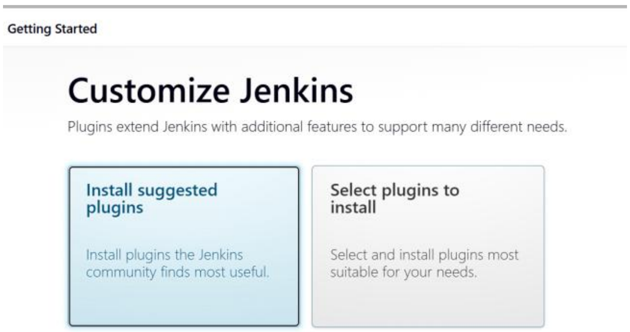
Step no:4 Paste in the console with Cat command to see the password.

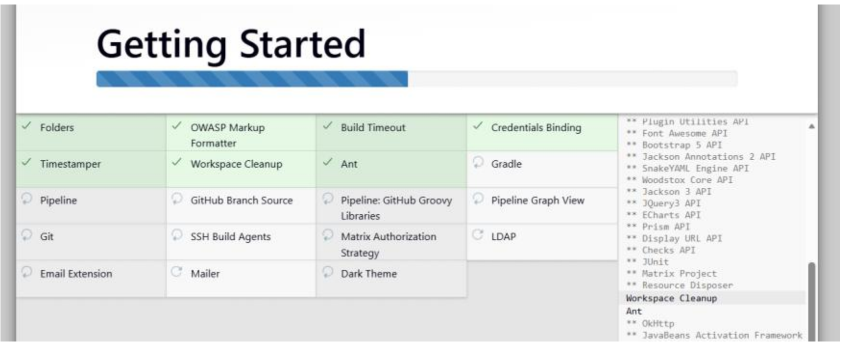


Now copy the password and paste it in the Jenkins login page password dialog box and login.

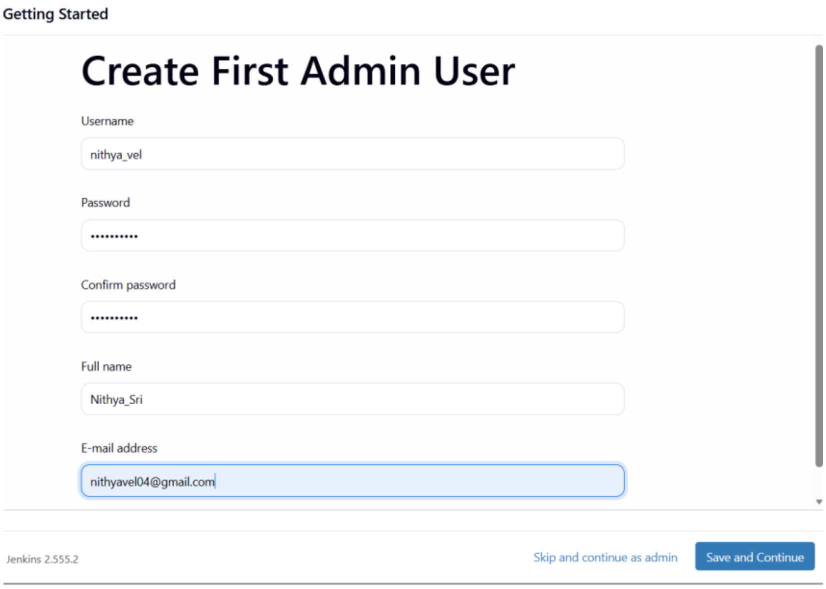


Step no:5 It will ask you to install the suggested plugins so that our app works fine without any problem. Install them.

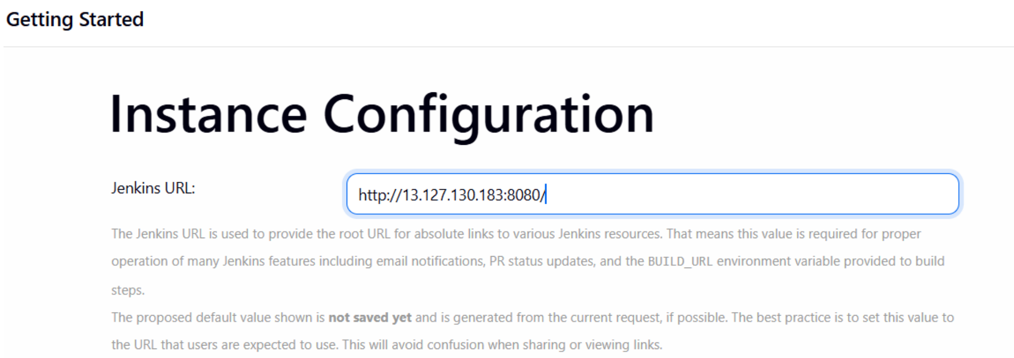




<u>Step no:6</u> Once the installation is done. The login page ask us to create a username password and our email address. Provide them save and continue.

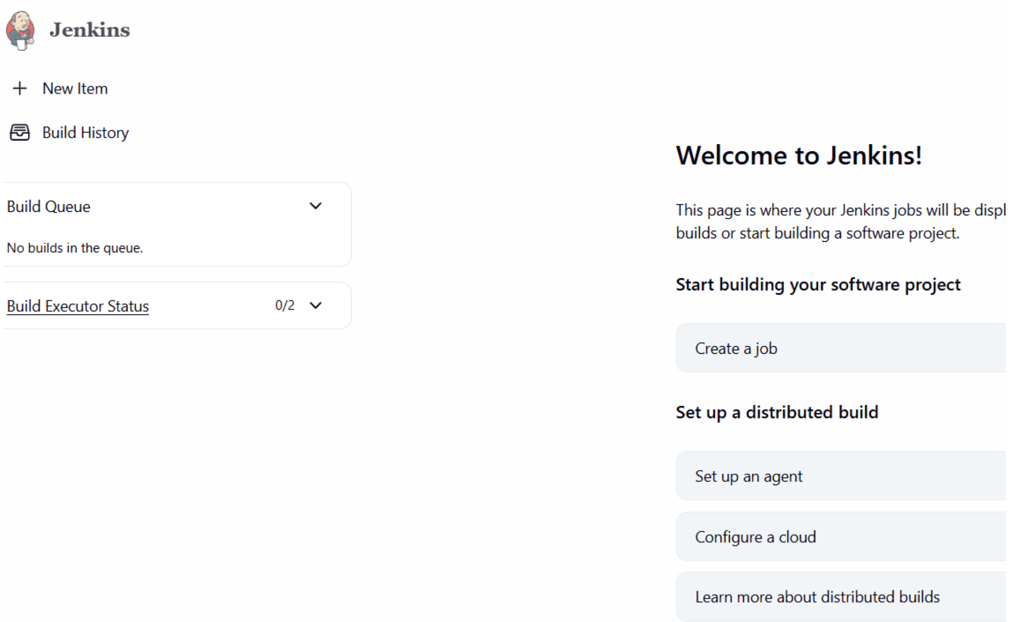


We can see the Jenkins url → click continue.

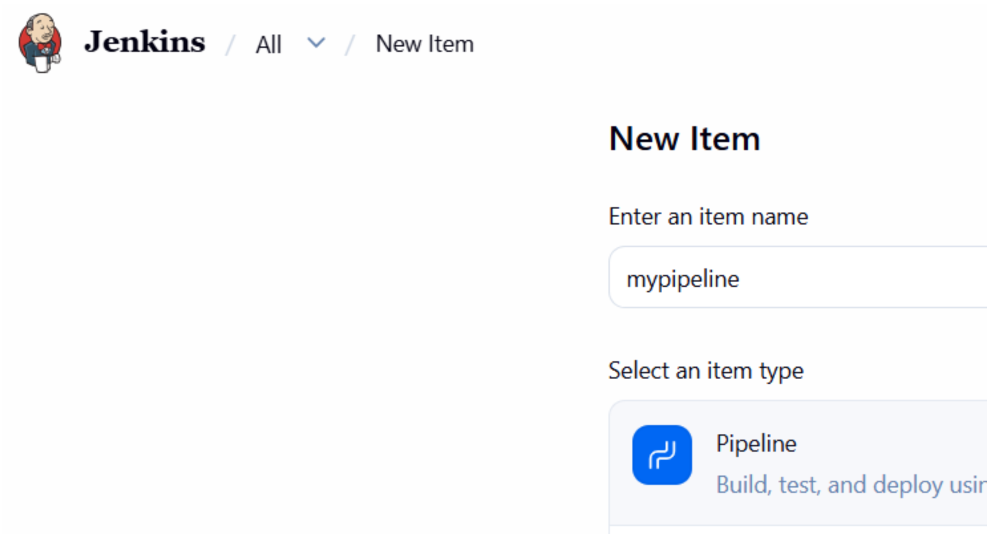


<u>Step no:7</u> We can see the Jenkins home page.

Click on new item



Name the pipeline as : **mypipeline** → type as: **Pipeline** → ok.



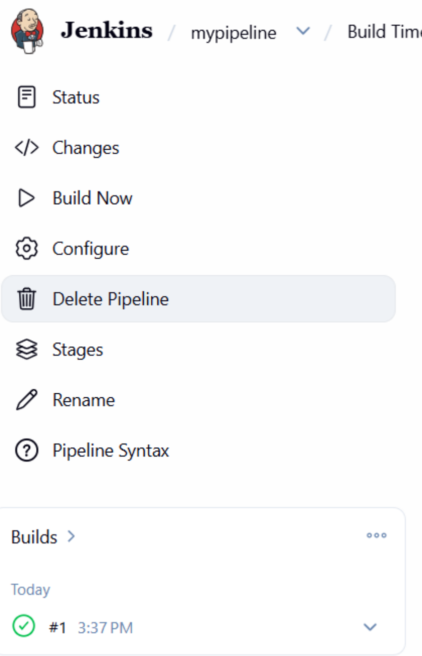
To visualise how the pipeline helps to build, test and deploy → scroll down to pipeline script.

We can write our own script or pull it from GitHub repo.

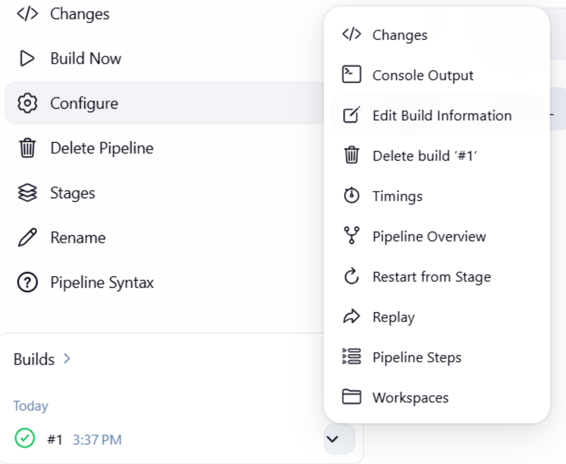
I am using a sample pipeline script provide by the Jenkins itself.

Choose save.

Step no:8 Now click Build now option. A build will be executed.



Once executed, we need to see the output of Jenkins script through **console output**.



We can see that our script has been executed successfully there.